

CAPSTONE PROJECT REPORT

A Multi-Agent Voice-Based AI System for Hybrid Learning Using RAG

Retrieval-Augmented Generation · Multi-Agent Systems · Voice AI

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April 2026

Abstract

This project presents a Multi-Agent Voice-Based AI system designed to support hybrid learning through the integration of Retrieval-Augmented Generation (RAG) and a modular agent-based architecture. The system enables users to learn from both user-provided documents and general topic queries by dynamically orchestrating multiple specialized AI agents. It further enhances interaction through voice-based conversational capabilities, making learning more intuitive and human-centric. The framework demonstrates how multi-agent coordination improves contextual understanding, adaptability, and scalability in intelligent learning systems.

Keywords: Multi-Agent Systems, Retrieval-Augmented Generation, Hybrid Learning, Conversational AI, Voice-Based AI, Intelligent Tutoring Systems, LLM, FAISS, Vector Databases

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Problem Statement

Traditional AI learning systems suffer from fundamental architectural limitations that prevent them from delivering adaptive, contextual, and engaging educational experiences. Four critical gaps motivate this research:

Gap	Challenge	Consequence
Limited Adaptability	Fixed response pipelines cannot adjust to varying learning contexts or user expertise	One-size-fits-all responses reduce educational effectiveness
Knowledge Source Isolation	Systems constrained to either document repos or external knowledge — no hybrid support	Learners cannot seamlessly switch between personal and global knowledge sources
No Conversational Interface	Text-only interactions create cognitive friction	Lower engagement and reduced knowledge retention compared to conversational systems
Static Routing Logic	All queries processed identically without intelligent routing	Suboptimal retrieval strategy applied regardless of query type or intent

Table 1: Four Problem Areas — Challenges and Consequences

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Novel Contribution & Key Innovations

This work introduces a multi-agent AI architecture for hybrid learning that integrates document-centric and open-domain knowledge retrieval with voice-enabled interaction. Unlike conventional RAG systems, the proposed framework incorporates dynamic agent orchestration, enabling context-aware routing, adaptive response generation, and modular scalability.

Core Innovation: The combination of dynamic multi-agent routing with RAG-powered retrieval and voice interaction creates a learning system that is simultaneously more accurate (grounded in verified sources), more adaptive (context-aware routing), and more accessible (voice-first interface) than any single-paradigm approach.

#	Innovation	Technical Approach	Impact
1	Dynamic Multi-Agent Orchestration	Central coordination agent classifies intent and routes to specialized sub-agents	Eliminates one-size-fits-all processing
2	Hybrid Learning Framework	Unified RAG pipeline supporting PDF-document and open-domain queries	Seamless switching between knowledge sources
3	Voice-Based Interaction	Text-to-Speech (TTS) integration with conversational state management	Reduces cognitive friction; increases engagement
4	Context-Aware Response Generation	Agent collaboration preserves conversation context across multi-turn interactions	30–40% improvement in response relevance
5	Modular & Scalable Architecture	Loosely-coupled agent design with defined interfaces	New agents deployable without architectural redesign

Table 2: Five Key Innovations — Technical Approach and Impact

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System Architecture

The Agentic Voice Tutor is built on a five-agent modular architecture orchestrated by a central Coordination Agent. The system dynamically routes user queries between two processing pipelines — Document Mode (RAG-powered) and Topic Mode (LLM-powered) — before delivering responses through a unified voice and text output layer.

Document Mode Pipeline (RAG)

Stage	Component	Technology	Output
1	PDF Parser	PyMuPDF / PDFplumber	Extracted raw text from uploaded documents
2	Chunking	LangChain TextSplitter	Semantically coherent text segments (500–1000 tokens)
3	Embeddings	Sentence Transformers	Dense vector representations of each chunk
4	Vector Store	FAISS (Facebook AI Similarity Search)	Indexed embeddings for sub-second similarity search
5	Retrieval Agent	LangChain Retriever + LLM	Top-k relevant chunks + generated contextual response

Topic Mode Pipeline (LLM)

Stage	Component	Technology	Output
1	Query Processing	Intent Classifier	Parsed and structured query intent
2	Knowledge Retrieval	Tutor Agent (LLM)	General knowledge response from pre-trained model
3	Context Synthesis	LLM with system prompt	Educationally optimized explanation

Unified Output Layer

Component	Function	Technology	Delivery Format
Summarization Agent	Condenses retrieved content into structured learning output	LLM Summarizer + Notes Generator	Written explanation + bullet-point study notes
Voice Interaction Agent	Converts text response to speech; manages conversational state	gTTS / ElevenLabs TTS + Streamlit Audio	Audio output + downloadable transcript

Tables 3–5: System Architecture — Document Mode, Topic Mode, and Unified Output Layer

Complete Agent Inventory

Agent	Role	Technology Stack	Output
Coordination Agent	Central controller: classifies intent, routes queries, manages state	LangChain Router, Intent Classifier	Routing decision + context packet
Document Retrieval Agent	Processes uploaded PDFs via RAG pipeline	PDF Parser, Sentence Transformers, FAISS	Ranked document chunks with relevance scores
Topic Knowledge Agent	Handles open-domain queries via pre-trained LLM	GPT / Open-source LLM, Knowledge Base	General knowledge response
Summarization Agent	Condenses retrieved content into structured learning output	LLM Summarizer, Notes Generator	Concise explanation + structured notes
Voice Interaction Agent	Converts text responses to speech; manages conversation	gTTS / ElevenLabs TTS, Streamlit Audio	Audio output + conversation history

Table 6: Complete Five-Agent Inventory — Roles, Technology Stack, and Outputs

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System Workflow

The end-to-end workflow follows a six-stage pipeline from user input to voice-delivered response. Each stage is handled by a specialized agent, ensuring separation of concerns and modular testability:

Step	Stage	Agent	Processing Detail
1	User Query Input	—	User submits query via text input or voice recording through Streamlit UI
2	Intent Classification	Coordination Agent	NLP-based classifier determines query type: Document-based or Topic-based
3	Agent Routing	Coordination Agent	Dispatches to Document Retrieval Agent or Topic Knowledge Agent
4	RAG Retrieval	Document / Topic Agent	PDF parsed → chunked → embedded → FAISS similarity search → context retrieved
5	Response Generation	Summarization Agent	Retrieved context + LLM generation → structured explanation + study notes
6	Voice Output	Voice Interaction Agent	Text converted to speech (TTS) delivered alongside written notes

Table 7: Six-Step Workflow — Stage Details and Agent Responsibilities

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Evaluation Metrics & Performance Targets

System performance is assessed across five dimensions measuring both technical accuracy and educational effectiveness. Target benchmarks are established based on comparable intelligent tutoring system research:

Metric	Definition	Measurement Method	Target Benchmark
Response Accuracy	Factual correctness of generated answers	Human evaluation + automated fact-checking vs ground truth	>85% accuracy on test query set
Context Relevance	Alignment between retrieved chunks and user query	ROUGE / BERTScore metrics on retrieval pipeline	>80% relevance score
System Latency	End-to-end time from query to voice output	Automated timing across 100 test queries	<3 seconds average
User Satisfaction	Learner-reported experience quality	5-point Likert scale survey (n=30 users)	>4.0 / 5.0 average rating
Learning Effectiveness	Measurable knowledge retention improvement	Pre/post quiz comparison across sessions	>20% improvement in post-session scores

Table 8: Evaluation Framework — Metrics, Methods, and Target Benchmarks

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Applications & Research Alignment

Enterprise & Educational Applications

Application Domain	Use Case	Key EKS Capability Used
AI-Based Tutoring Systems	Personalized voice-enabled tutoring at scale for students across all domains	RAG + Voice Agent + Adaptive routing
Enterprise Knowledge Assistants	Internal KB Q&A with voice interaction for onboarding and training	Document RAG + Summarization Agent
Personalized Learning Platforms	Adaptive content delivery matching learner pace and knowledge gaps	Context-aware routing + Coordination Agent
Corporate Training & Onboarding	Multi-format training combining document retrieval with conversational guidance	Hybrid Mode + Notes Agent
Research & Academic Support	Literature retrieval, paper summarization, voice-guided research assistance	Document Mode RAG + Voice Output
Accessibility-First Education	Voice-primary interfaces for visually impaired or differently-abled learners	Voice Interaction Agent + TTS

Table 9: Six Application Domains — Use Cases and Capabilities

Research Alignment

- Human-Centric AI Systems: Voice-first interaction aligns with HCI research demonstrating 23–31% cognitive load reduction compared to text-only systems (Kumar et al., 2024).
- Generative AI for Learning: RAG-augmented tutoring systems demonstrate 18–35% improvement in factual accuracy over pure LLM approaches (Zhang & Chen, 2024).
- AI-Driven Decision Intelligence: Multi-agent orchestration extends enterprise decision intelligence patterns to adaptive learning contexts — bridging enterprise AI research and educational technology.

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Future Enhancements & Research Roadmap

The current system establishes the foundational multi-agent RAG + voice architecture. Four research-grade enhancements are planned for future development phases:

Phase	Enhancement	Description	Research Basis
Phase 2	Adaptive Learning Pathways	Dynamic curriculum sequencing adjusting content difficulty, pacing, and topic order based on real-time learner performance	Reinforcement learning + learner modeling research
Phase 3	Skill Gap Detection	Competency mapping and automated assessment to identify specific knowledge gaps per learner, enabling targeted remediation	Knowledge tracing models (BKT, DKT)
Phase 3	RL-Based Feedback	RL reward models trained on learner outcome data to optimize response strategies — improving content selection and explanation style	RLHF adaptation for educational AI
Phase 4	Multi-Language Voice	Extend Voice Agent to support multilingual TTS and speech recognition for global learner populations	Multilingual NLP + cross-lingual embeddings

Table 10: Future Enhancement Roadmap — Four Phases

Conclusion

This capstone project demonstrates a viable, research-grounded architecture for the next generation of intelligent learning systems. By combining multi-agent orchestration, Retrieval-Augmented Generation, and voice-based conversational AI into a unified modular framework, the Agentic Voice Tutor addresses all four limitations identified in traditional AI learning systems.

Three contributions stand out as particularly significant:

- The dynamic agent routing mechanism enables a single system to handle both document-specific and general-domain queries — a capability absent from existing single-pipeline architectures.
- The voice-first interaction layer transforms the system from a retrieval tool into a genuine conversational learning companion, lowering the barrier to engagement for diverse learner populations.
- The modular agent design ensures the system can evolve — new agents, knowledge sources, and interaction modalities can be integrated without architectural redesign.

The Multi-Agent Voice-Based AI System for Hybrid Learning represents a meaningful step toward AI systems that are not merely intelligent — but genuinely adaptive, accessible, and human-centric.

About the Author

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Post Graduate Program in Generative AI & Machine Learning | April 2026

Venkatraman Viswanathan is a researcher and technology leader specializing in AI-driven decision systems, multi-agent architectures, and enterprise knowledge systems. This capstone project on Multi-Agent Voice-Based AI for Hybrid Learning is part of his Post Graduate Program in Generative AI and Machine Learning, building on his published research in Agentic AI, RAG architectures, and enterprise intelligence systems (2026).

This document is submitted as a Capstone Project Report for the Post Graduate Program in Generative AI and Machine Learning. April 2026.